

Tutorial on Reverse Engineering of Gene Regulatory Networks

Dr. Roberto Pagliarini, *University of Udine*

Dr. Grete Francesca Privitera, *University of Catania*

Dr. Dora Tortarolo, *University of Turin*



CELLS: THE BUILDING BLOCKS OF LIFE

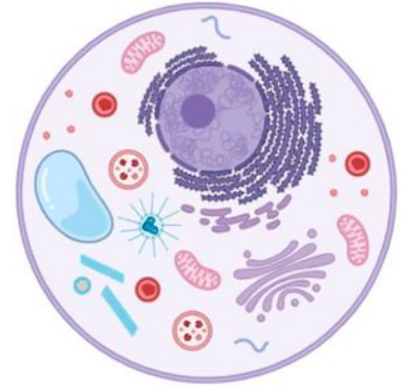
- Cell Membrane
- Cytoplasm
- Genetic Material (DNA)

CELLS: THE BUILDING BLOCKS OF LIFE

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Prokaryote (e.g. *Escherichia Coli*)



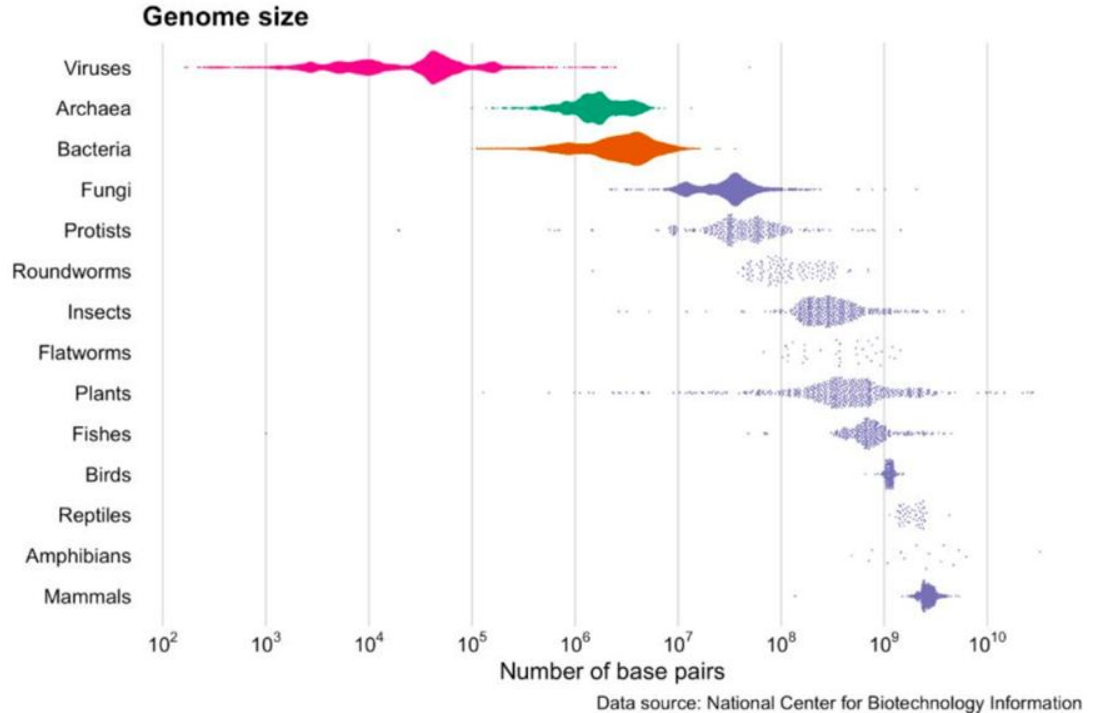
Eukaryote (e.g. *human fibroblast*)

	E. Coli	Human
Cell volume	1 μm^3	10,000 μm^3
Proteins/cell	$\sim 4 \times 10^6$	$\sim 4 \times 10^{10}$
Size of genome	4.6×10^6 bp 4500 genes	3.2×10^9 bp 21,000 genes

THE PARADOX OF COMPLEXITY

C-value: total amount of DNA in the haploid genome of an organism, measured in picograms or million base pairs

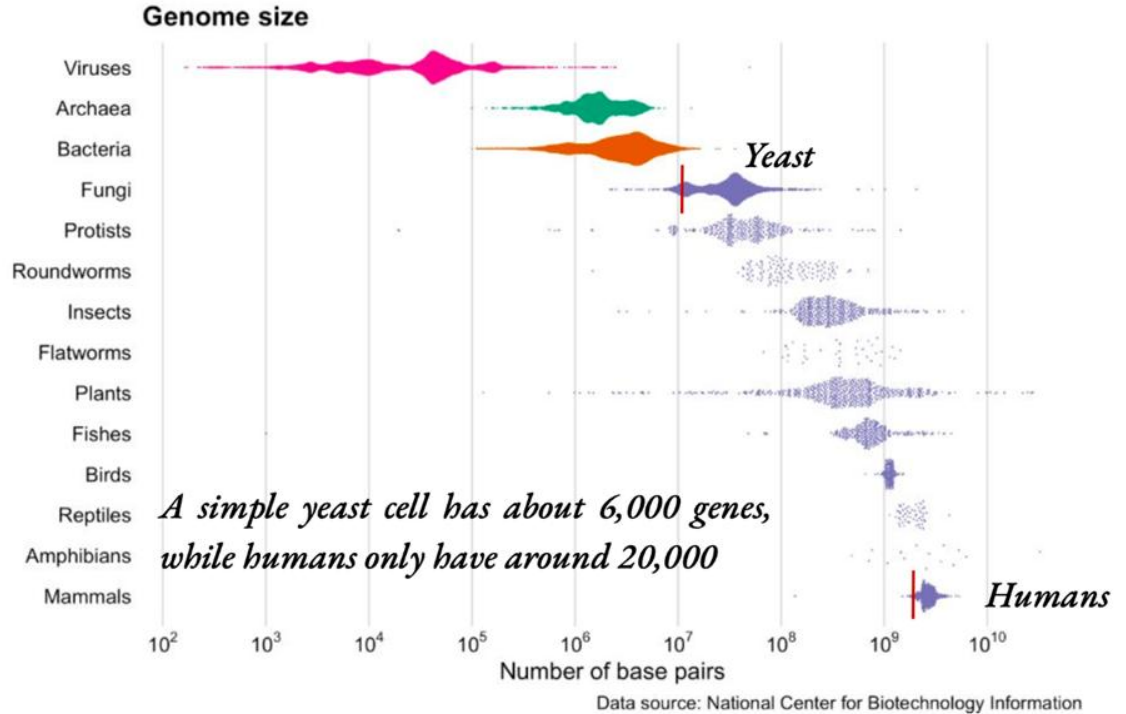
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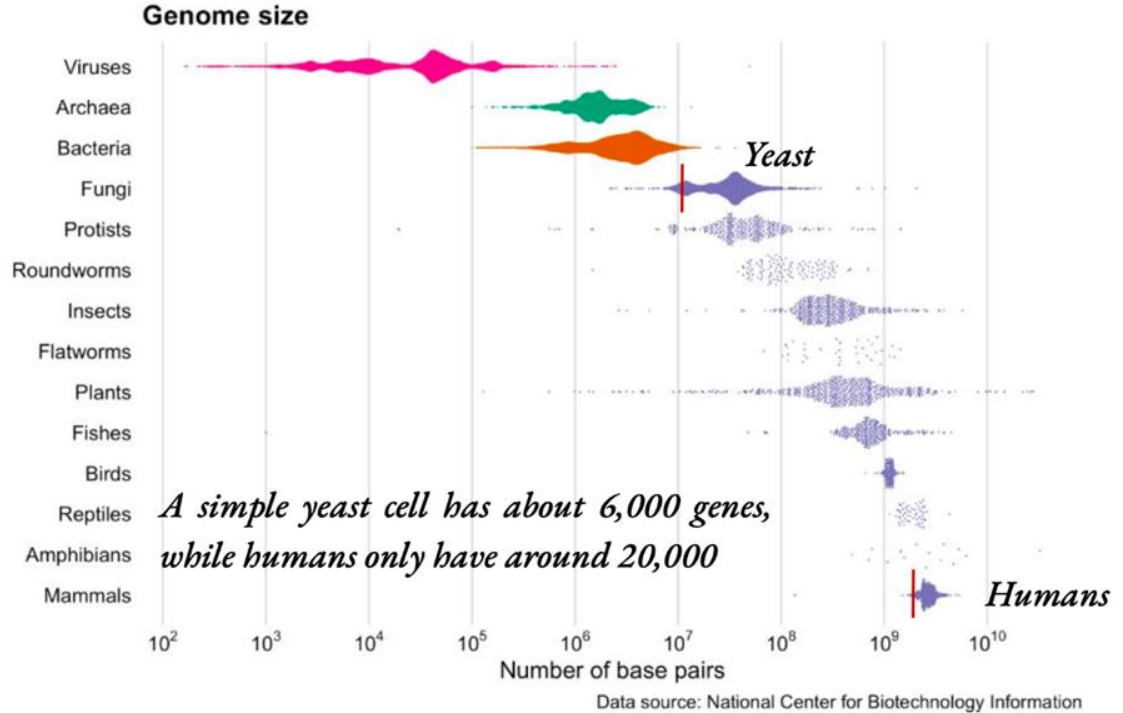
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Non - coding DNA

The extraordinary diversity of specialised cells is obtained through an **accurate control** of the regulation and expression of the genes.

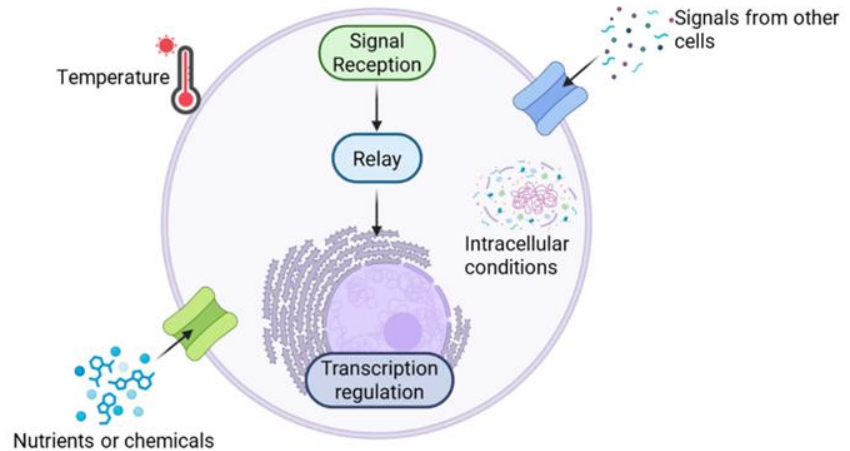


CELLS AS INFORMATION PROCESSORS

Signals interpretation

- *physical parameters (temperature and osmotic pressure)*
- *biological signaling molecules from other cells*
- *beneficial nutrients*
- *harmful chemicals*
- *the internal state of the cell*

Transcription networks: allow cells to determine the rate of production of each protein as well as the amount needed.

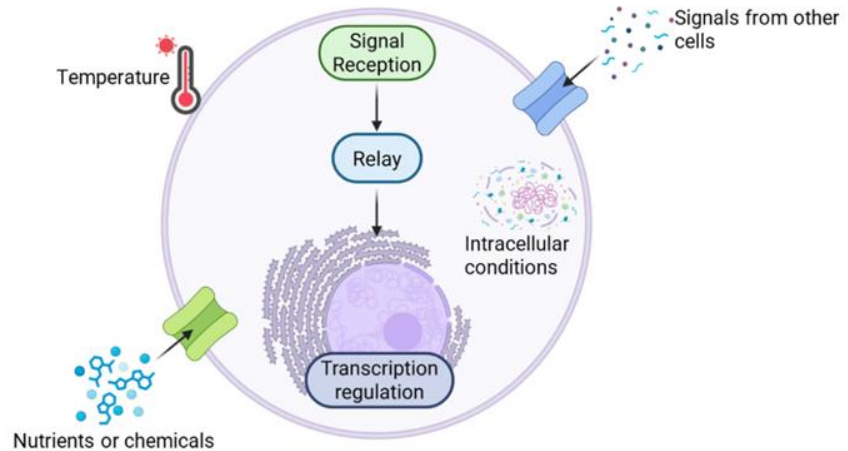


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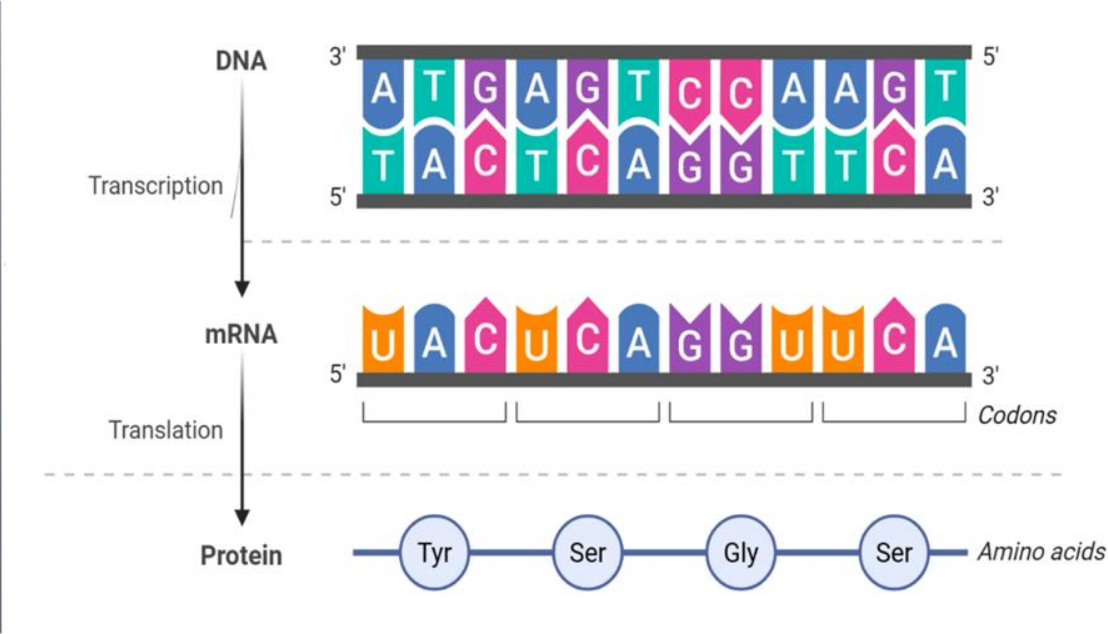
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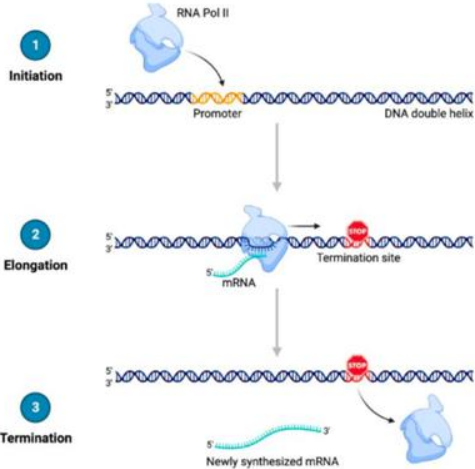


Cells must avoid energy waste

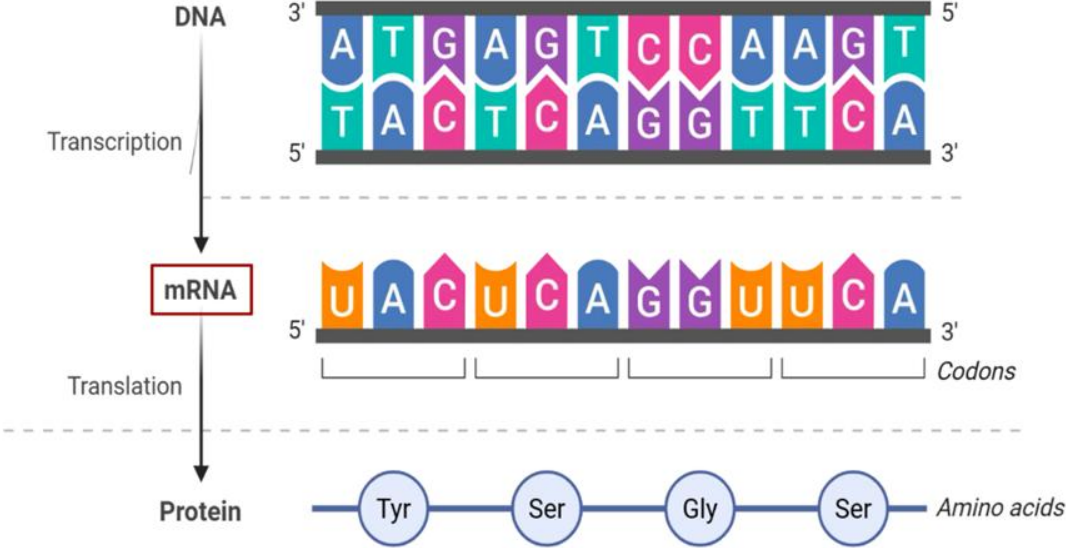
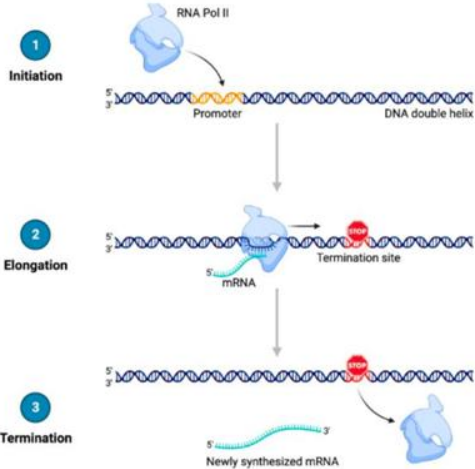
THE CENTRAL DOGMA OF BIOLOGY



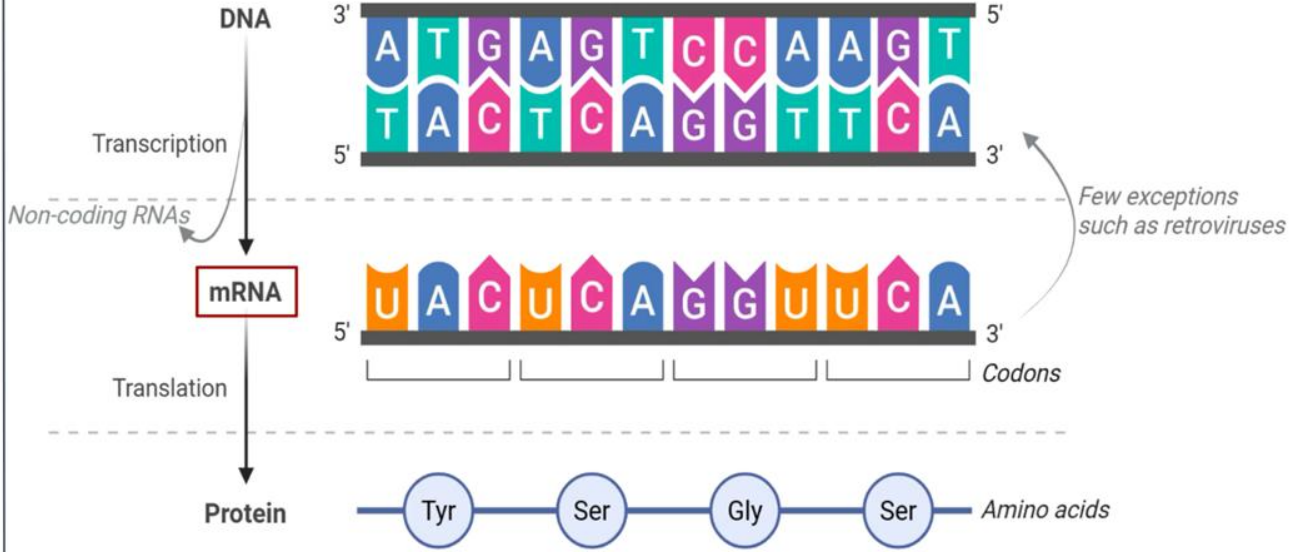
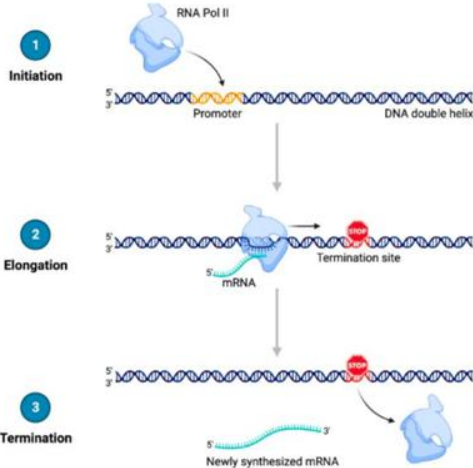
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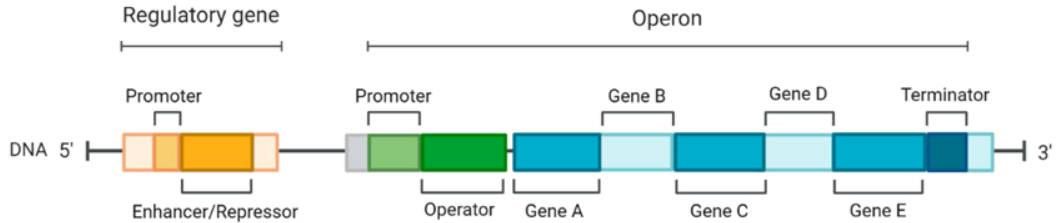
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THE BIOLOGICAL MACHINERY

- RNA polymerase
- Gene's structure
- Transcription factors (TFs)

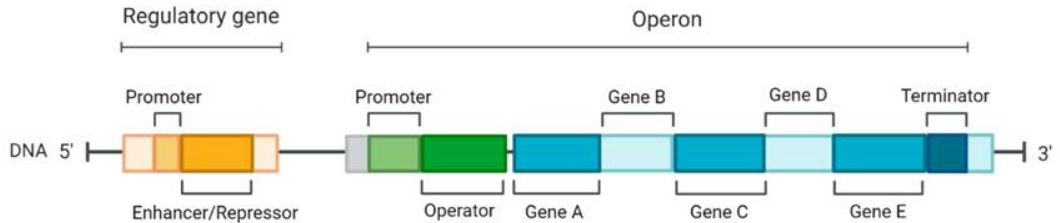
Prokaryotic gene structure



THE BIOLOGICAL MACHINERY

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Prokaryotic gene structure



Cis acting elements: promoter and control regions on DNA sequence

Trans acting elements: transcription factor

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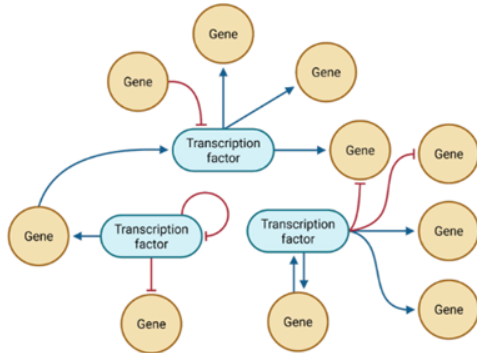


Temporal control (*E.Coli* example)

Action	Time
Binding of a small molecule signal to a TF, causing a change in its activity	~1 msec
Active TF binding to DNA site	~1 sec
Transcription and translation of a gene	~ minutes
Change of 50% in concentration of the protein	~ 1 hour

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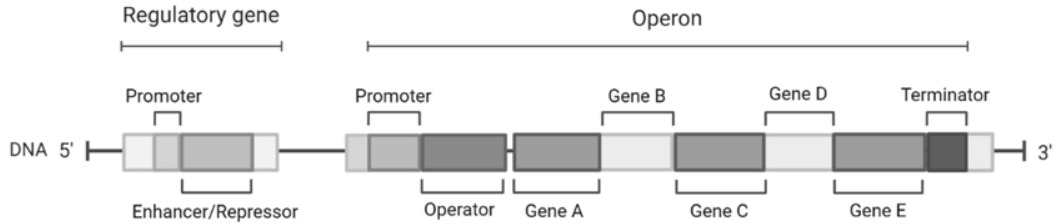
Combinatorial control

- TFs act together like **computational logic gates**. This allows the cell to integrate multiple different environmental signals before making a decision
- **Mixing and matching** just ~1,500 TFs allows precise control over ~20,000 genes in humans

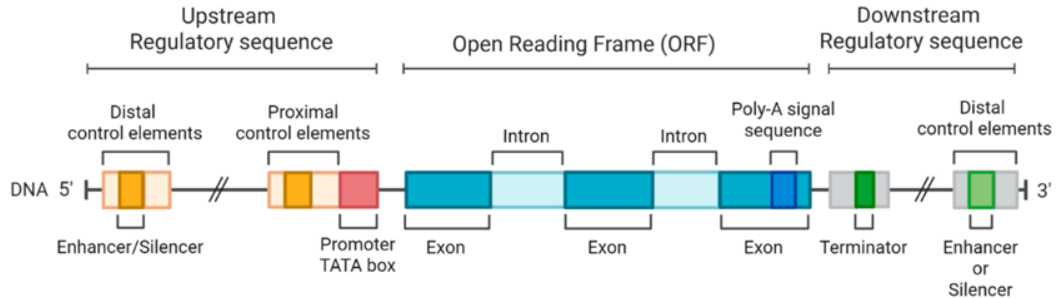
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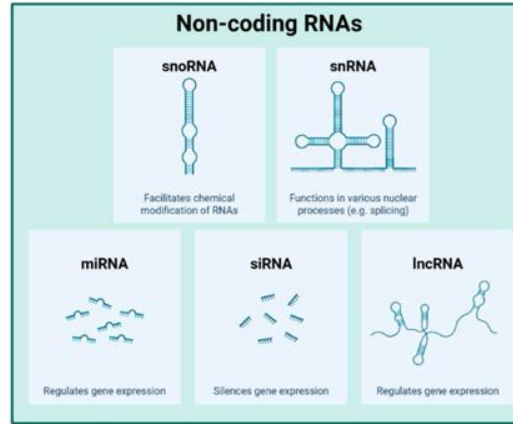
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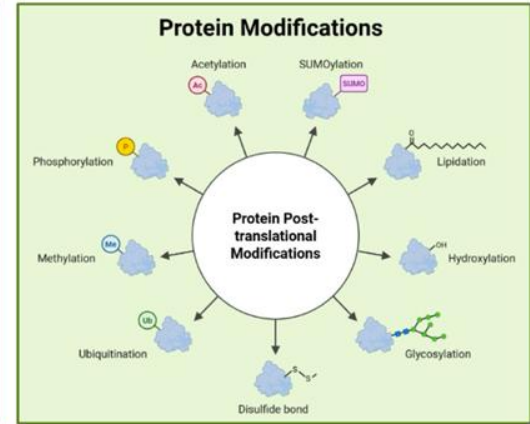
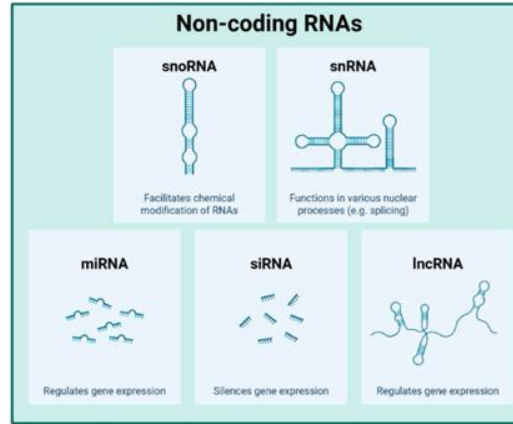
Eukaryotic gene structure



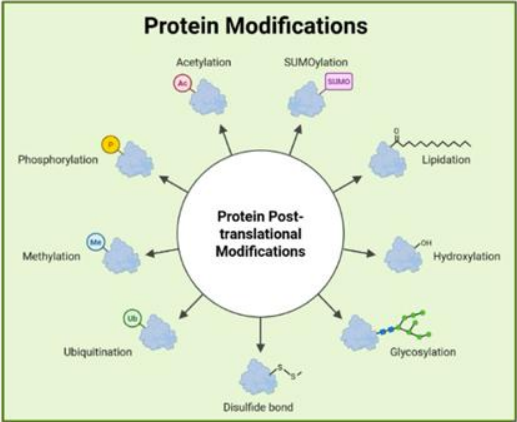
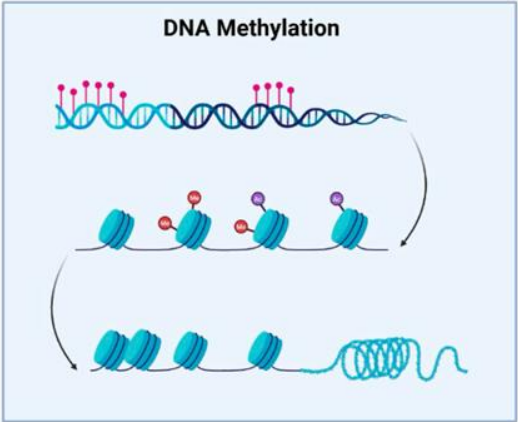
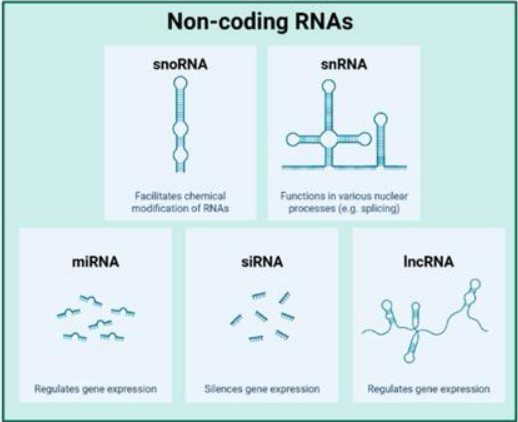
SCALING UP: Multi-Layered Regulation



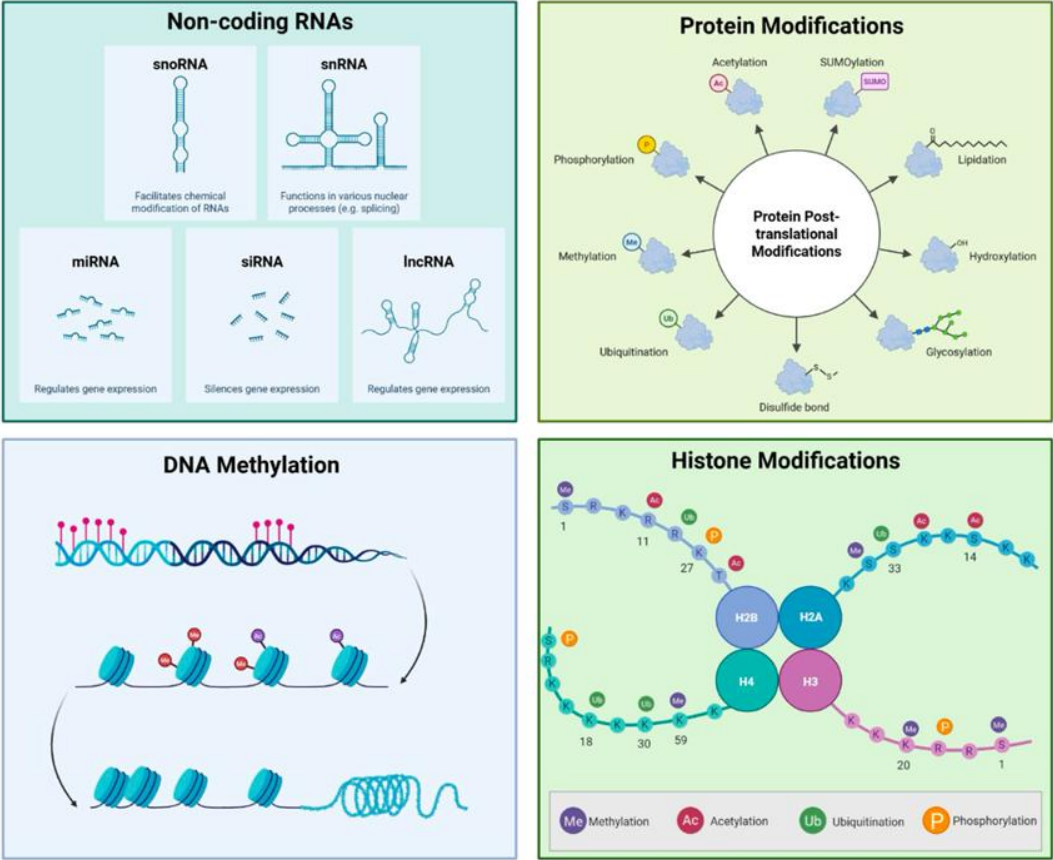
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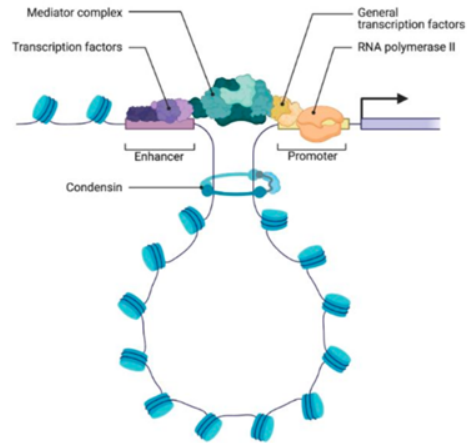


SCALING UP: Multi-Layered Regulation

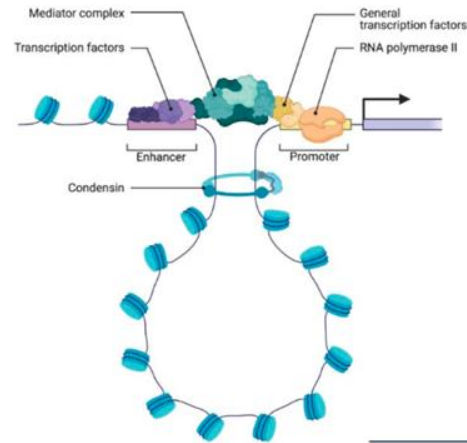


Adapted from Kim, S. (2026); BioRender (2026) BioRender.

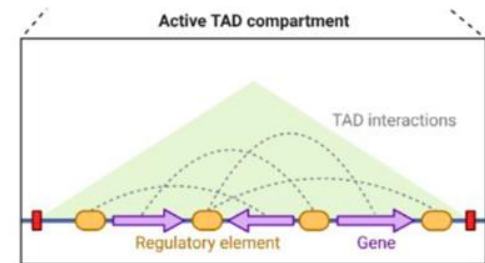
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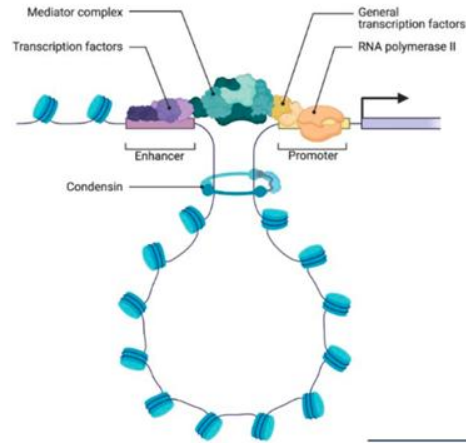


Topologically Associating Domains
TADs

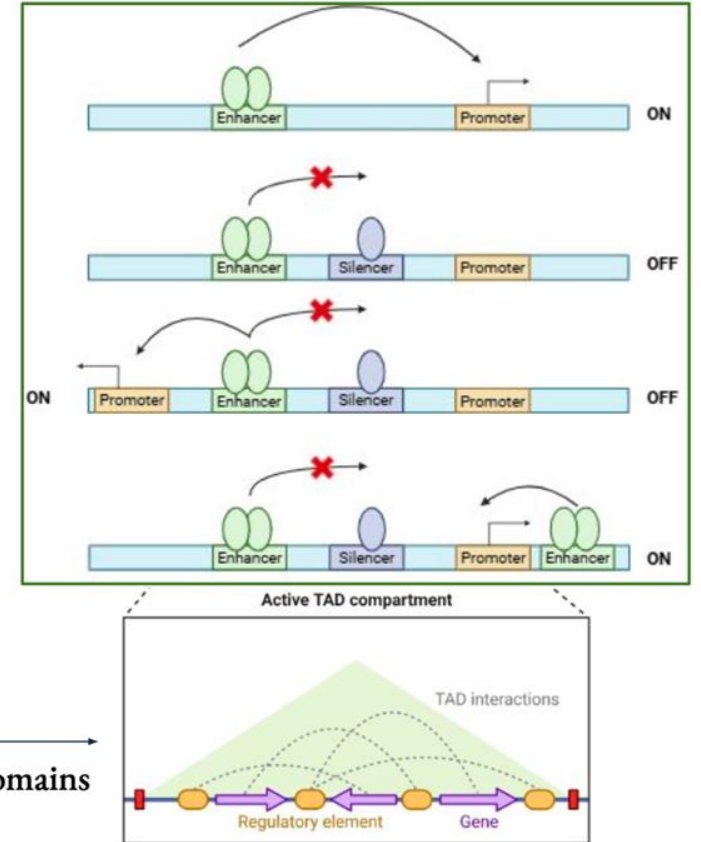


Zheng, H., et al., 2019, Nat Rev Mol Cell; Matharu N, et al., 2015, PLoS Genet

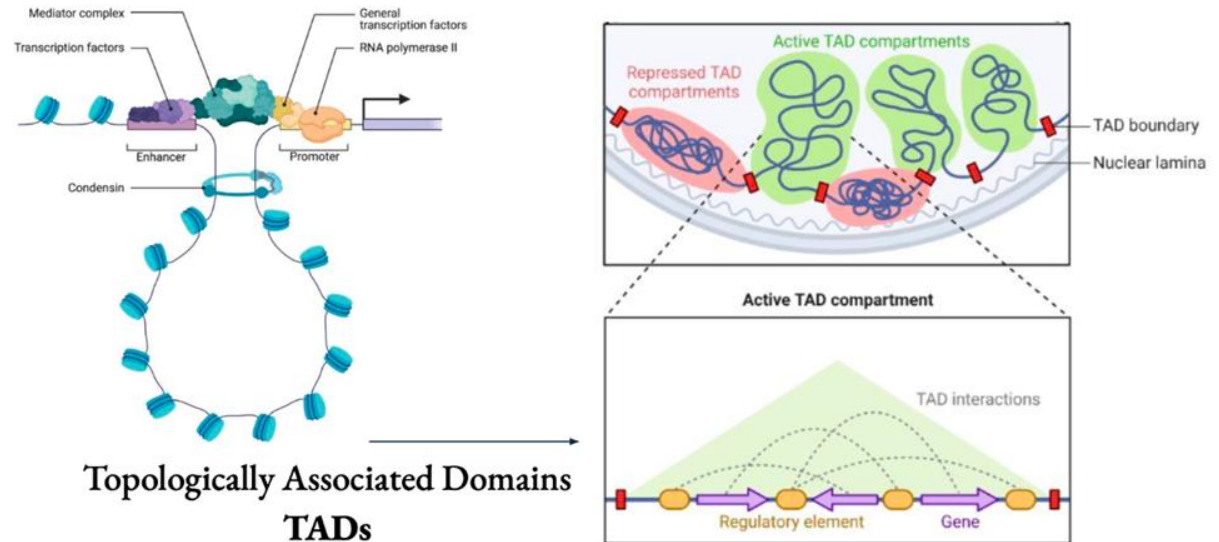
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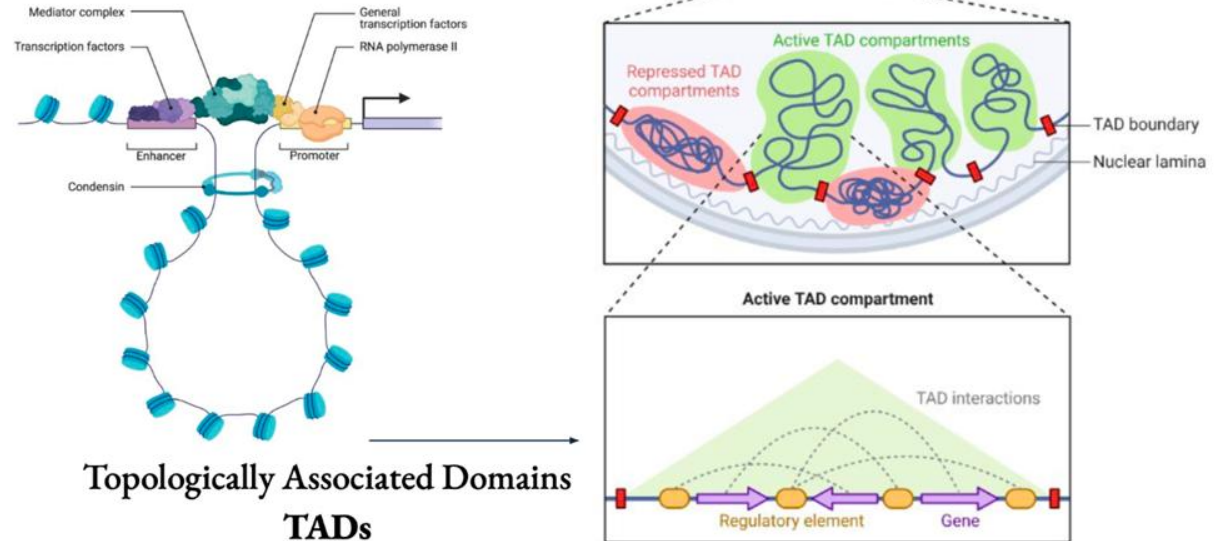
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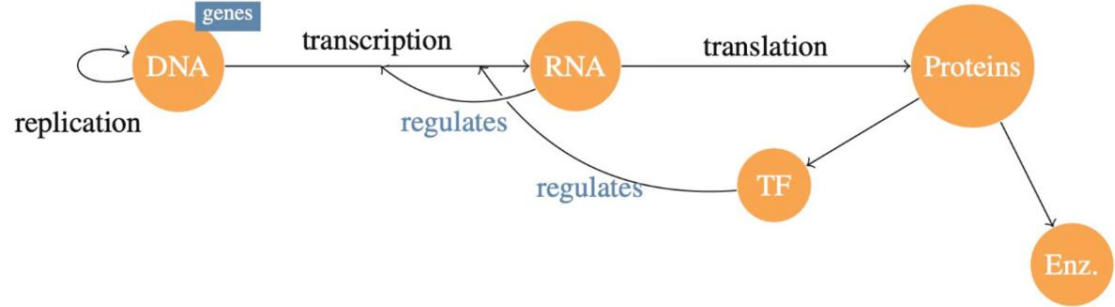
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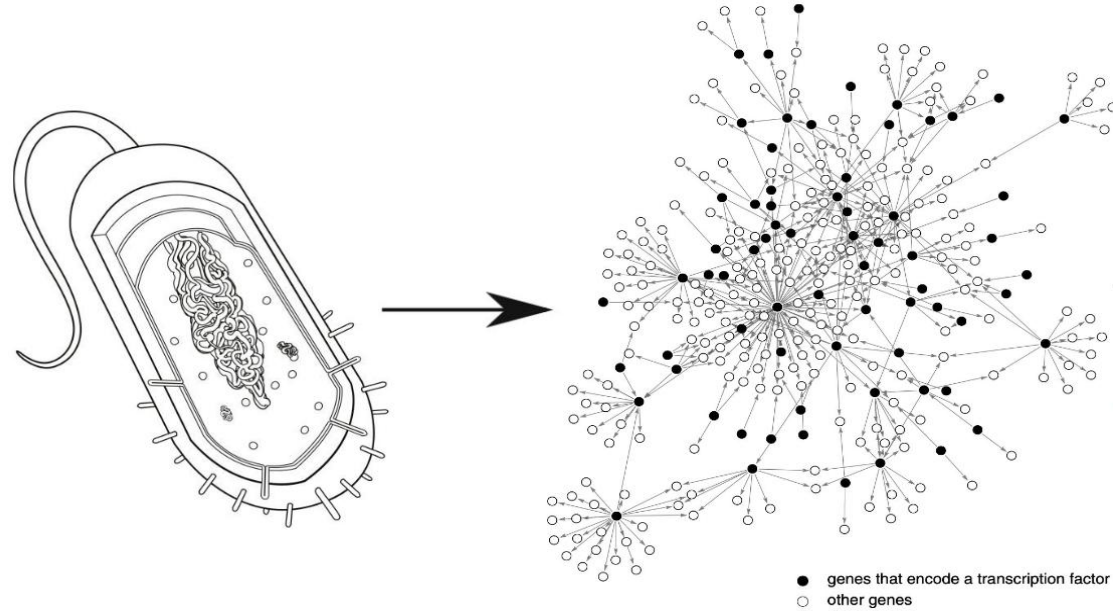
In eukaryotes, this involves a "committee" of proteins

- **activators,**
- **repressors,**
- **coactivators,**
- **chromatin-remodeling complexes**



Example of a real Gene Regulatory Network (GRN)

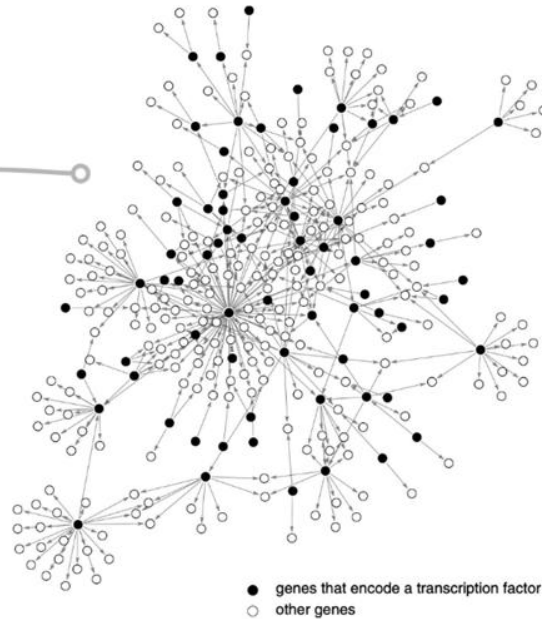
Transcription network of
Escherichia coli



Modelling Constitutes a Challenging Problem

Computational Task

Properly translate molecular biology knowledge into a **consistent** and **predictive GRNs model**

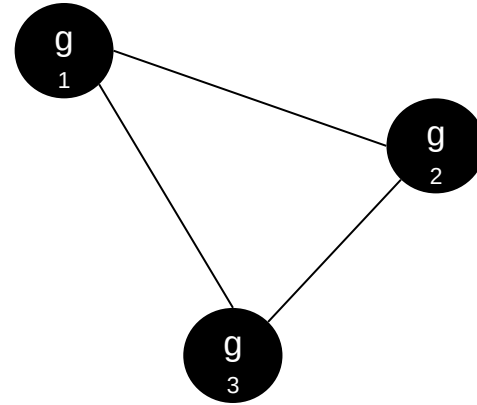


Alon U. (2019). *An Introduction to Systems Biology*. CRC Press.

GENE NETWORKS

A gene network is a **graph** used to represent *knowledge* about the relationship between biological entities

- intuitive
- quantitative
- behave as mathematical objects



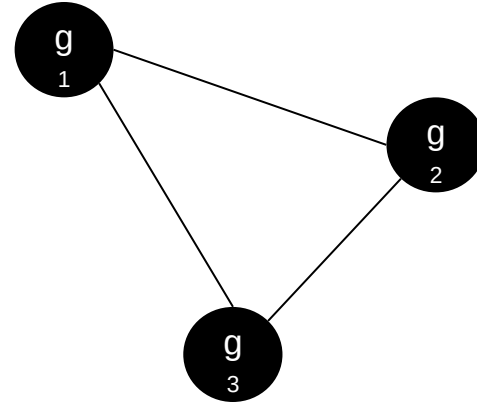
GENE NETWORKS

Nodes

A single *entity* in the network, for examples genes.

Edges

A *connection* between two nodes,
representing the relationship between nodes
It varies depending on the type of network.



GENE NETWORKS

Nodes

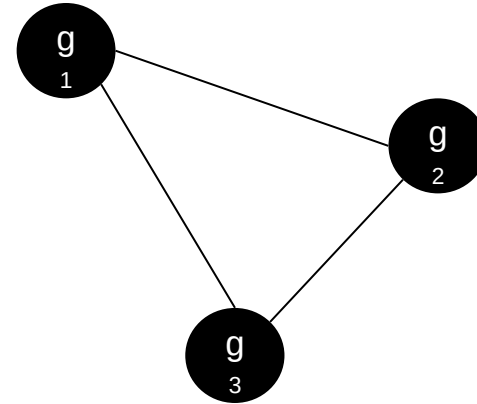
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Types of networks:

- Co-expression networks
- Gene regulatory networks (GRNs)



GENE NETWORKS

Nodes

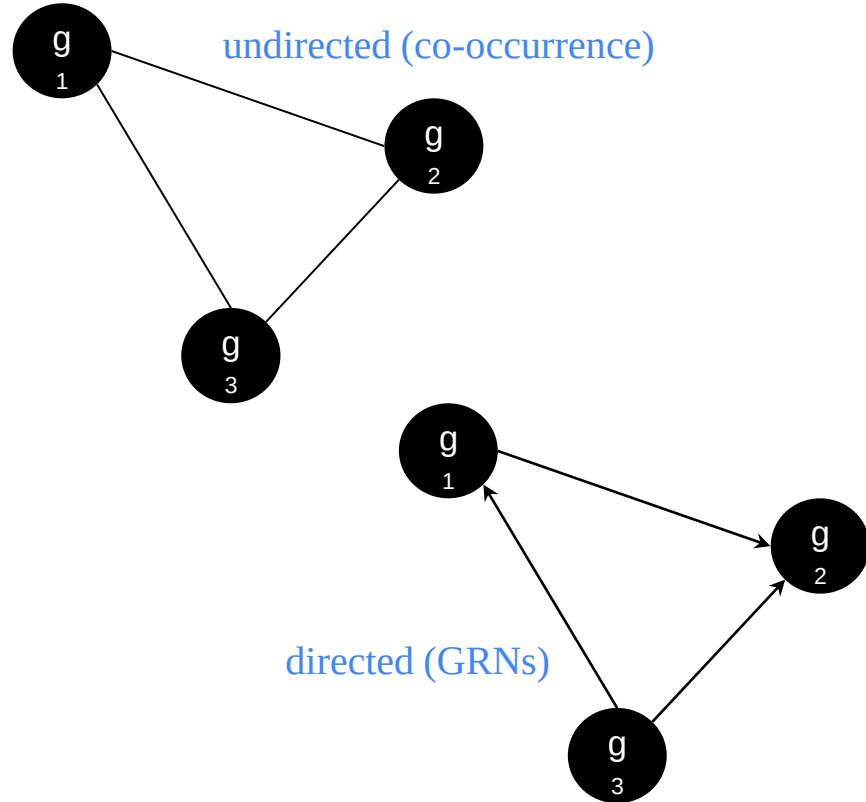
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Types of edges:

- directed vs undirected



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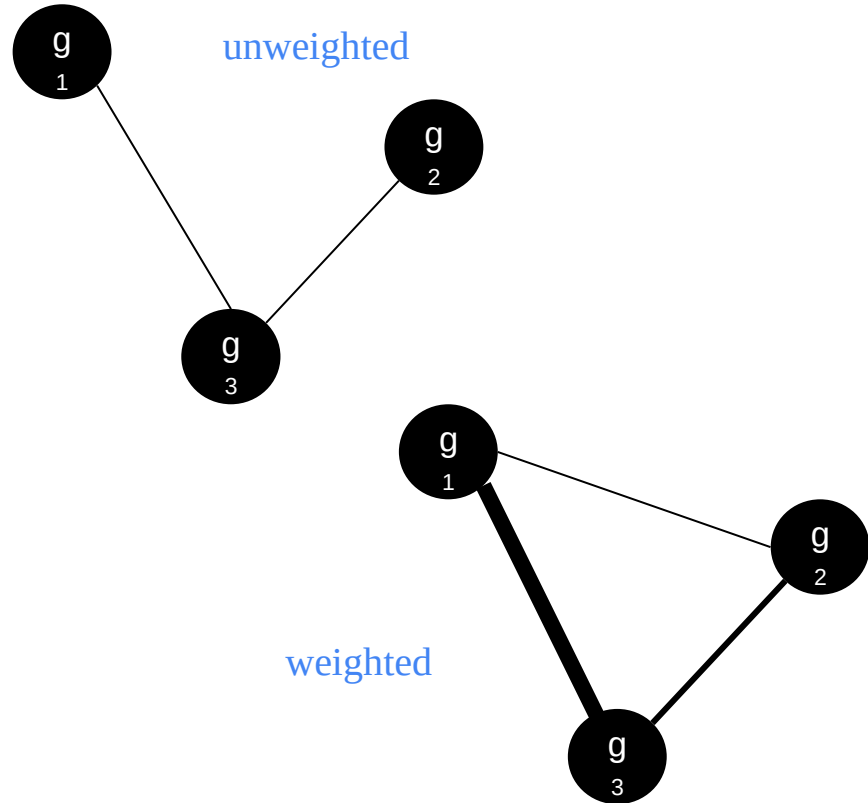
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Types of edges:

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- weighted vs unweighted



GENE NETWORKS

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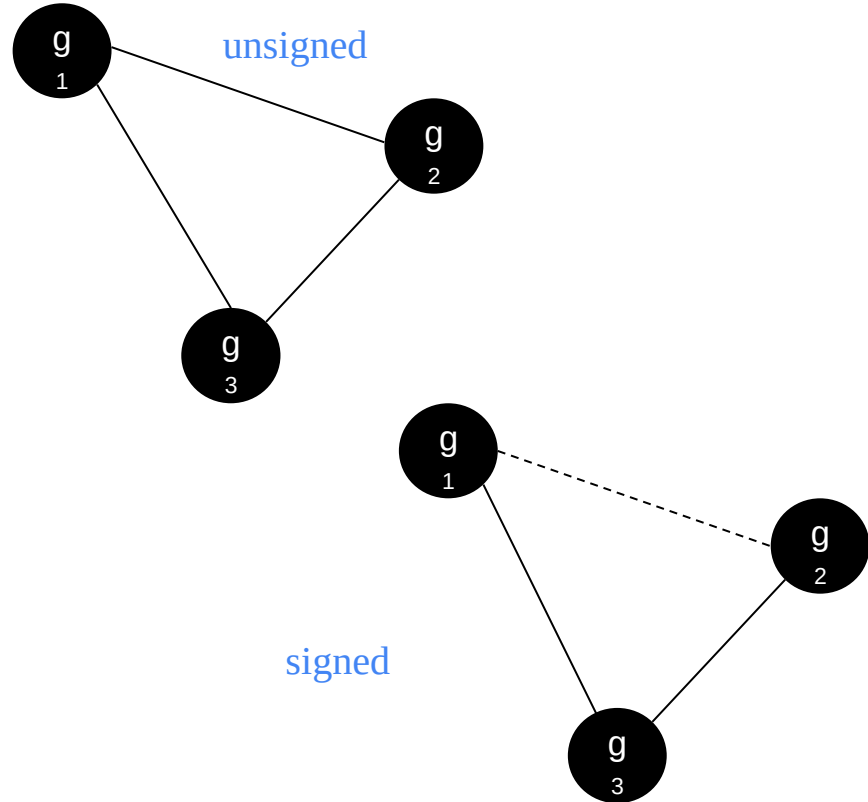
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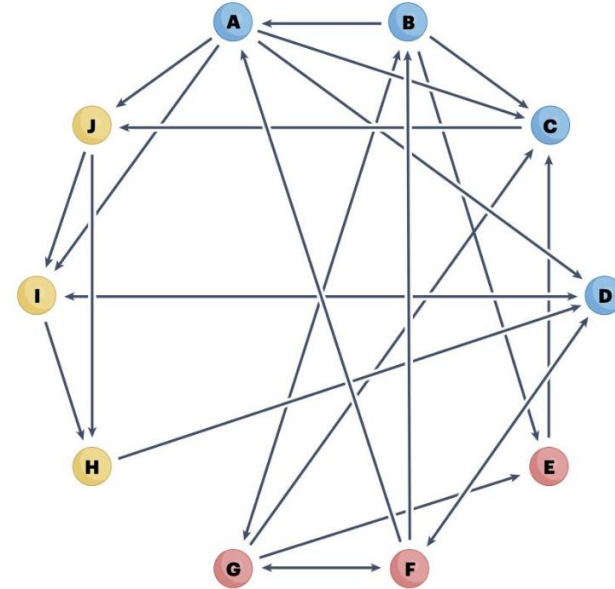
Types of edges:

- directed vs undirected
- weighted vs unweighted
- signed



CONSTRUCTING GENE NETWORKS

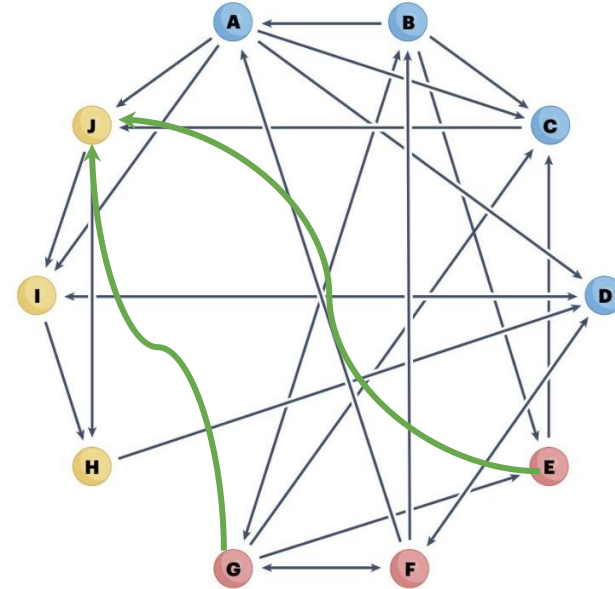
- biological knowledge from existing data
- *ad hoc* experiments to perform **genetic perturbations** to systematically identify **regulatory genes** and **elements**



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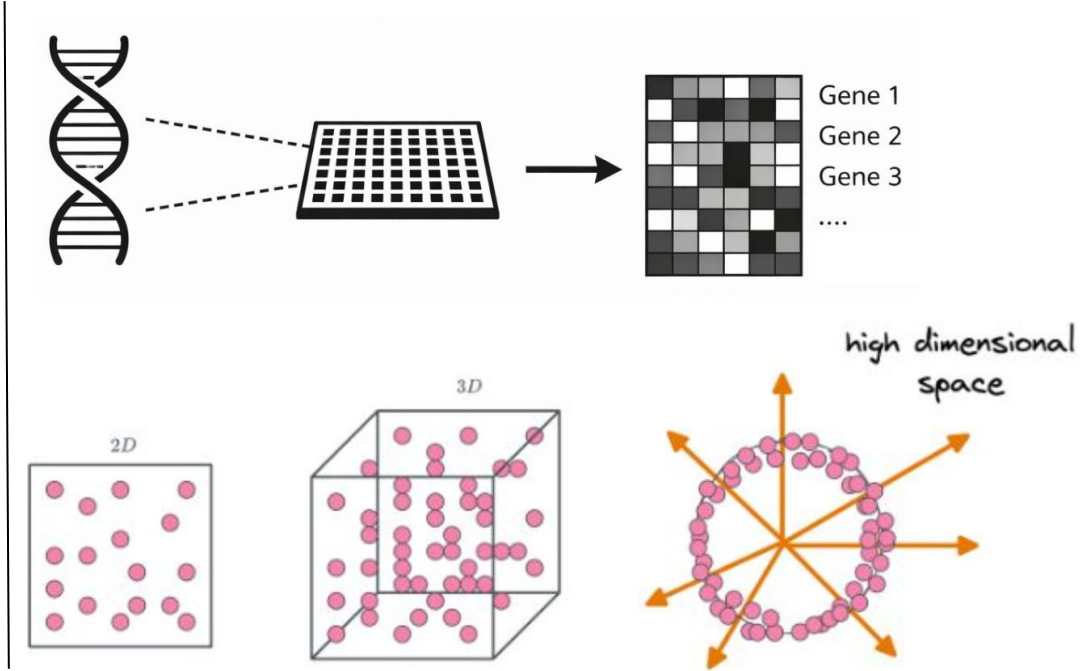
However, gaps in our knowledge require the development of **data-driven inferential methods** for GRNs reconstruction.



LIMITATIONS OF GNRs:

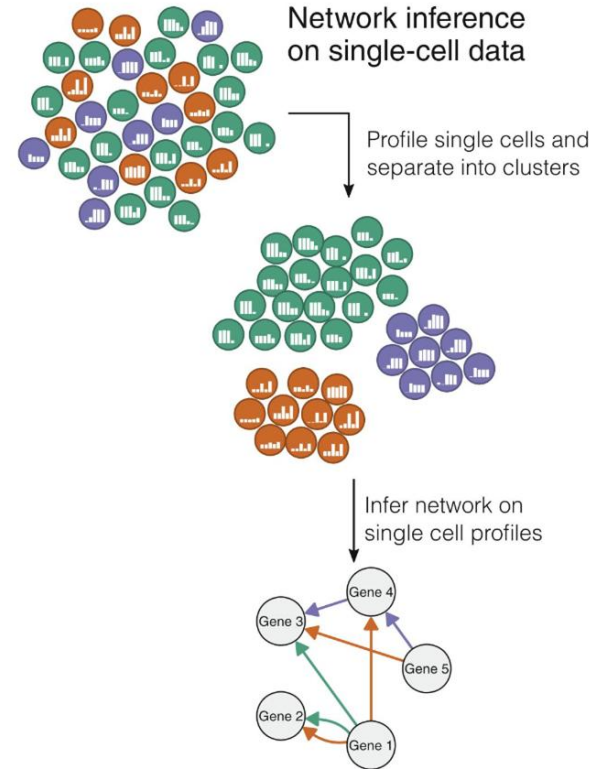
- **high dimensionality** (large p , small n)

*This effect is referred as the “**curse of dimensionality**”*



LIMITATIONS OF GNRs:

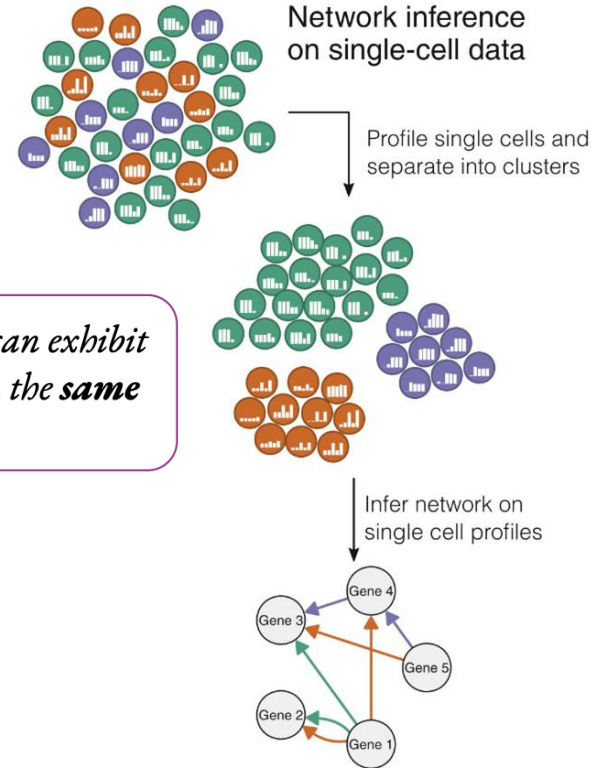
- ***High dimensionality*** (large p , small n)
- ***Noise***
 - *Technical noise*
 - *Biological noise*



LIMITATIONS OF GNRs:

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 - Biological noise

*Even cells that are **genetically identical** can exhibit a degree of **variability** in their responses in the **same environment***

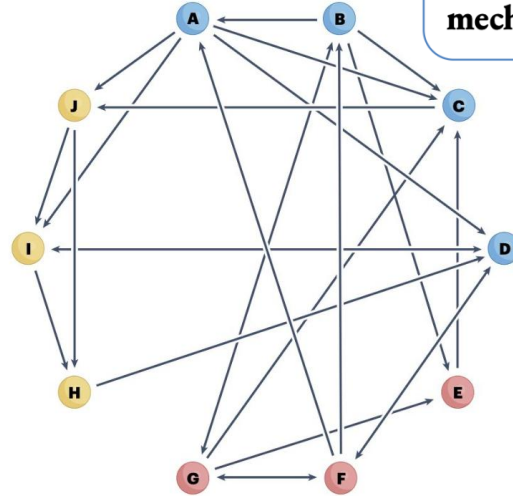


LIMITATIONS OF GNRs:

- **High dimensionality** (large p , small n)
- **Noise**
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- **Indirect effects**

*Many correlations **could result** significant **just by chance***

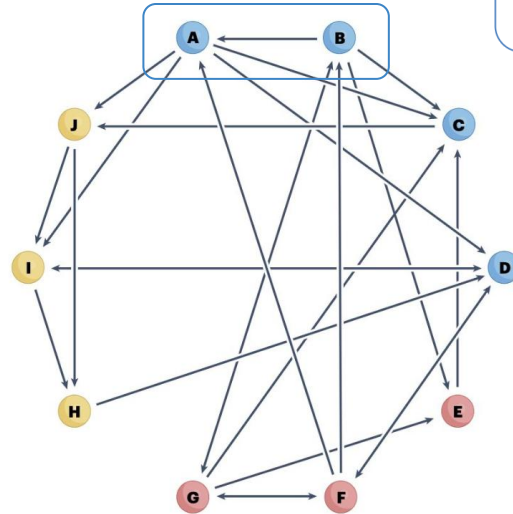
Leading to inferred networks which connections that can have a **poor mechanistic meaning**



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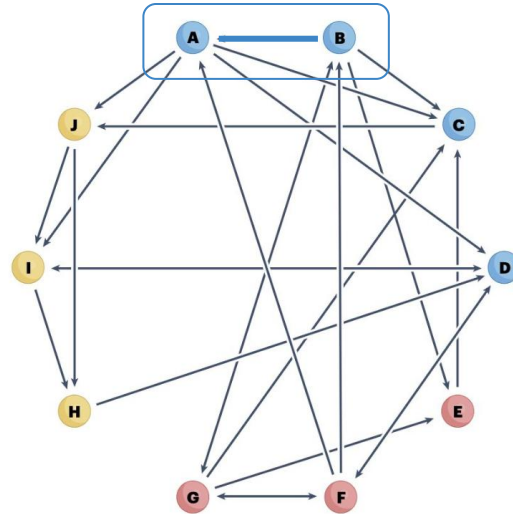


If nodes A and B result as highly correlated then a link is inferred connected them

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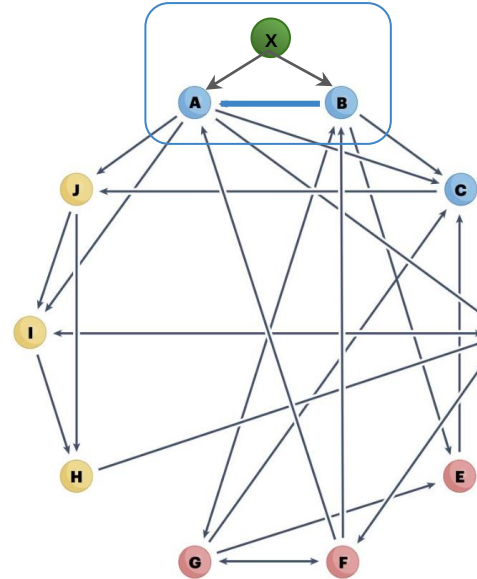
- **A directly regulates B**

This is a true regulatory edge

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- **Indirect regulation** by a common upstream regulator

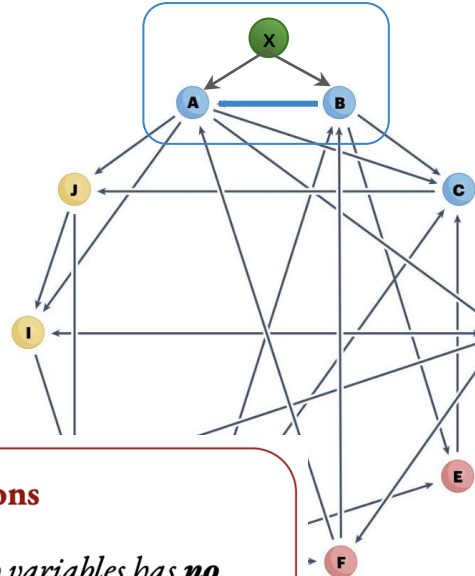
They are correlated not because they interact directly

LIMITATIONS OF GNRs:

- **High dimensionality** (large p , small n)
- **Noise**
 - Technical noise
 - Biological noise
- **Indirect effects**

With a low number of samples we are in **trouble to correct for multiple comparisons**

Many correlations could result significant just by chance



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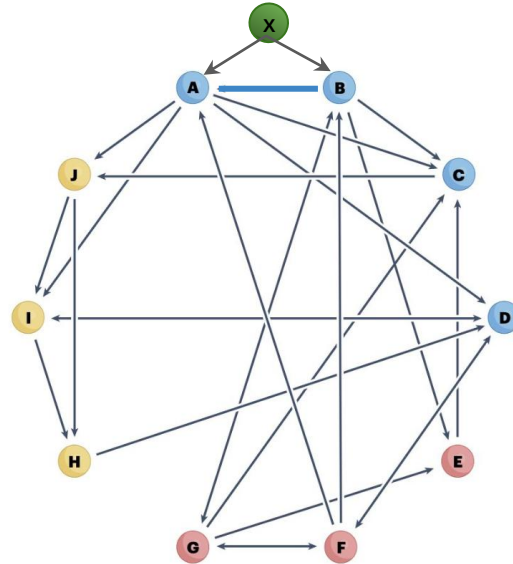
↑ **Spurious Correlations**

*Association between two variables has **no direct causal or mechanistic relationship***

LIMITATIONS OF GNRs:

- **High dimensionality** (large p , small n)
- **Noise**
 - *Technical noise*
 - *Biological noise*
- **Indirect effects**
- **Difficult to validate**

*Many correlations **could** result significant **just by chance***



FROM RAW COUNTS TO A CLEAN EXPRESSION MATRIX

1



INSPECTION

look at the data

2



NORMALIZATION

make samples comparable

3



TRANSFORMATION

stabilize variance

4



FILTERING

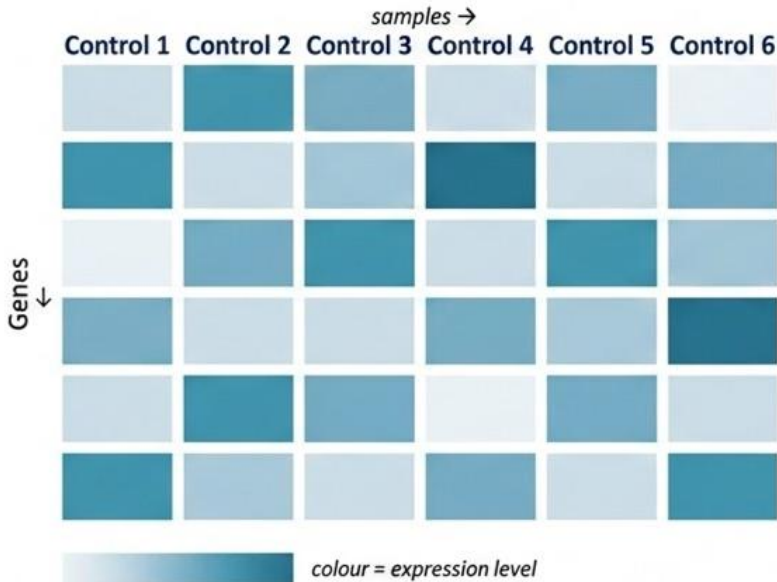
remove noisy genes

5



EXPLORATORY ANALYSIS

understand structure



Sample Metadata

Sample ID	Experimental Condition
Control 1	Control
Control 2	Control
Control 3	Control
Control 4	Control
Control 5	Control
Control 6	Control

Data inspection

1

Dimensions

How many genes, how many samples? Does it match the design?

2

Metadata

Condition, group and batch correctly assigned to each sample.

3

Missing / duplicates

NA values, duplicated genes, samples with few detected genes.

4

Distributions

Histograms / boxplots per sample reveal skew and outliers.

Inspection is quick, but it catches the problems that silently corrupt every later step.

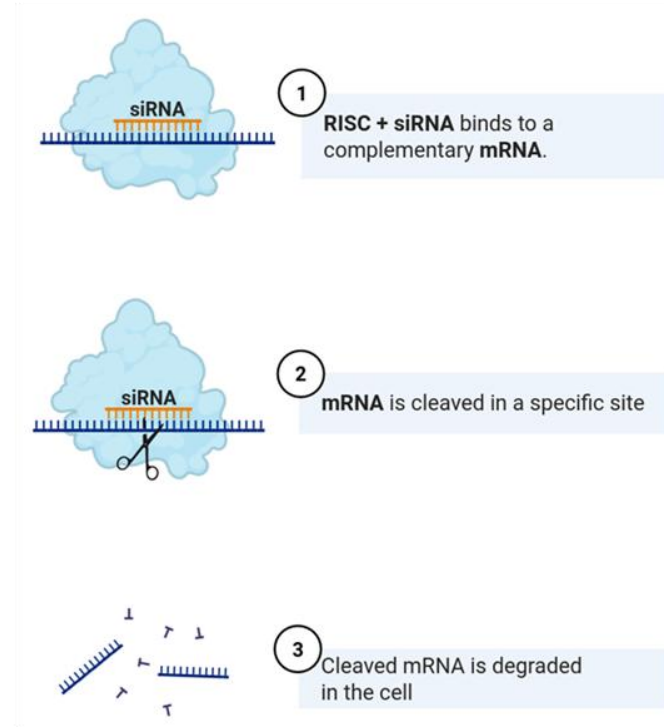
Types of Samples

Perturbation Strategy

- **Knockout (KO):** Complete gene removal at DNA levels (e.g. CRISPR)
- **Knockdown(KD):** Partial reduction in gene expression (e.g. RNAi)

Temporal Resolution

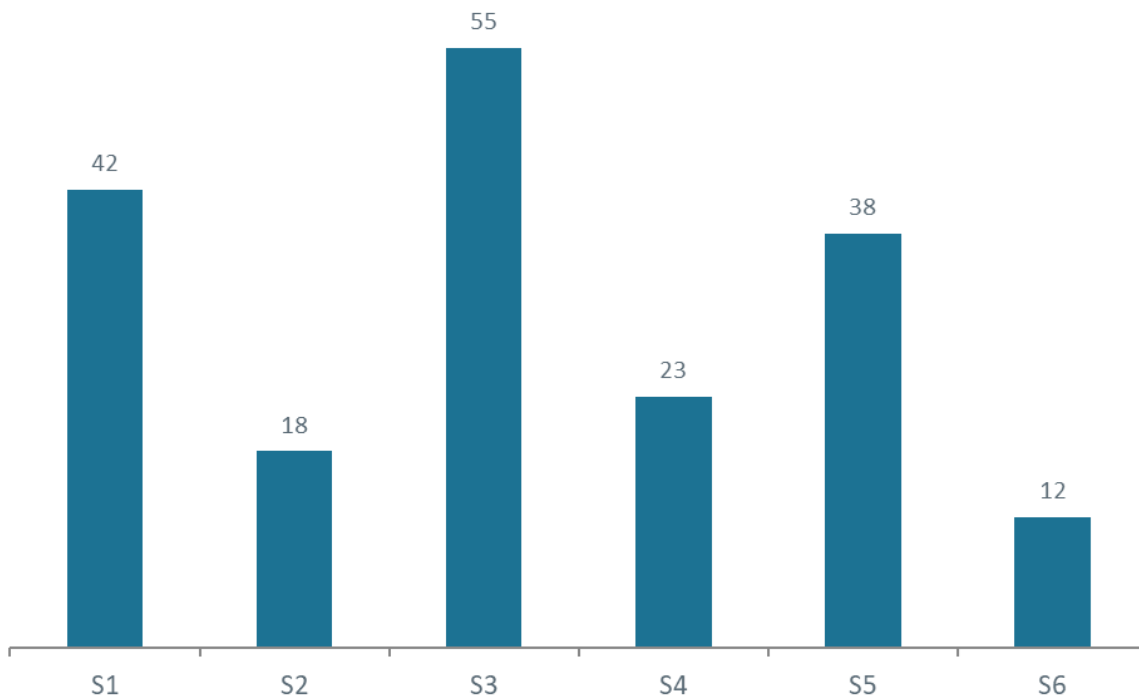
- **Steady state:** A "snapshot" of a biological state comparing the differences between two conditions
- **Time series:** To track expression trajectories with multiple time points



Normalization

Making samples comparable across an experiment

Total reads per sample (millions)



Objective of normalization

- Remove technical bias
- Make the samples comparable

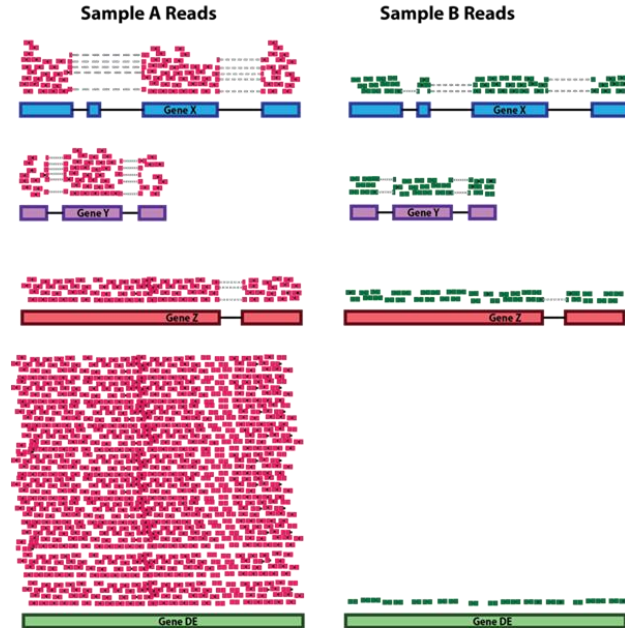
The bias comes from

- **Sequencing depth** — total reads differ between samples
- **Library composition** — a few genes can dominate the counts
- **Gene length** — longer genes attract more reads

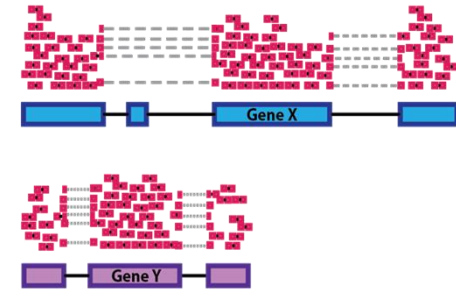
Why it's necessary: samples have different read counts and highly expressed genes distort the distribution.

Normalization

Sample B has overall **lower** sequencing depth compared to sample A



Sample A Reads



Few highly expressed genes make up a large portion of the total reads in Sample A compared to B

Normalization

The main methods

CPM

Counts per million

Formula

- $\text{Counts} / \text{Total reads} \times 10^6$

Pro

- Easy to compute

Con

- Does not correct composition

TPM

Transcripts per million

Advantages

- Comparable across samples
- More interpretable than RPKM

Use

- Exploratory analyses

RPKM / FPKM

Length-aware methods

Account for

- Gene length
- Sequencing depth

Limitation

- Not comparable across samples

TMM (edgeR)

Trimmed mean of M-values

Features

- Reference-based comparison
- Outlier exclusion

Pro

- Robust
- Standard in edgeR

Upper Quartile

Based on the 75th percentile

Pro

- Simple
- More robust than CPM

Con

- Less accurate than TMM / RLE

RLE (DESeq2)

Relative Log Expression

Features

- Median of ratios
- Assumes few differential genes

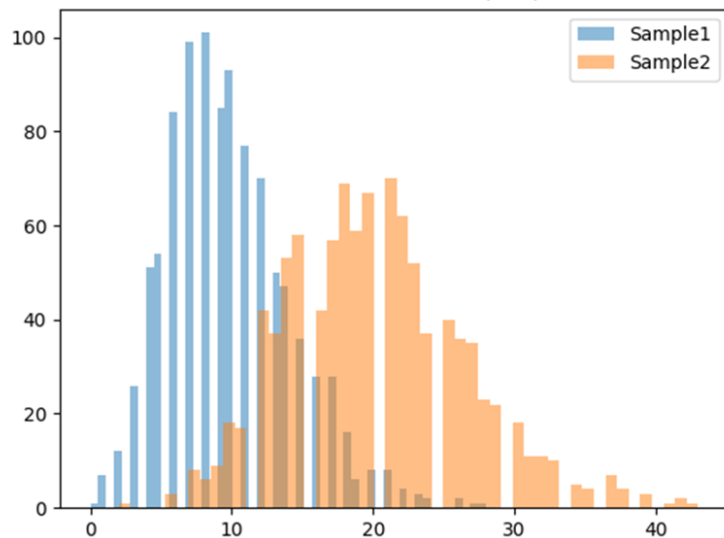
Pro

- Very stable
- Default in DESeq2

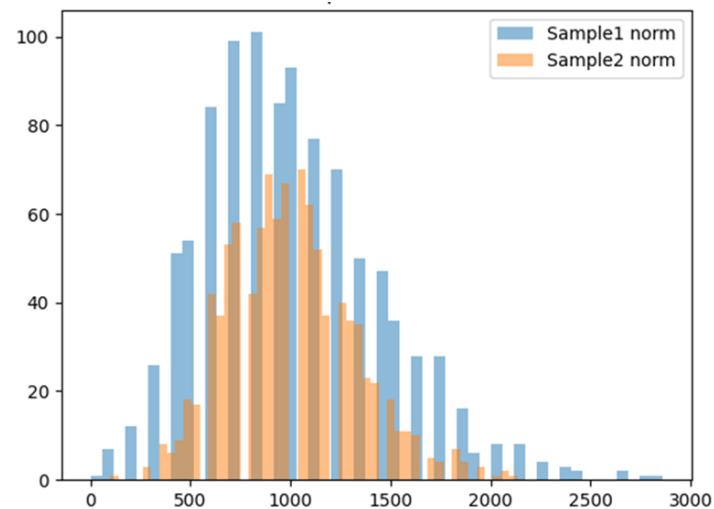
There is not an “universal” method. Usually RLE and TMM are used for the differential analysis and TPM for genes comparison!

Normalization

Raw counts distribution



Distribution after normalization (CPM)



Transformation

From skewed counts to constant variance

Counts data are **heteroscedastic**:
variance grows with the mean



Transformation achieves **homoscedasticity**:
constant variance across expression levels.



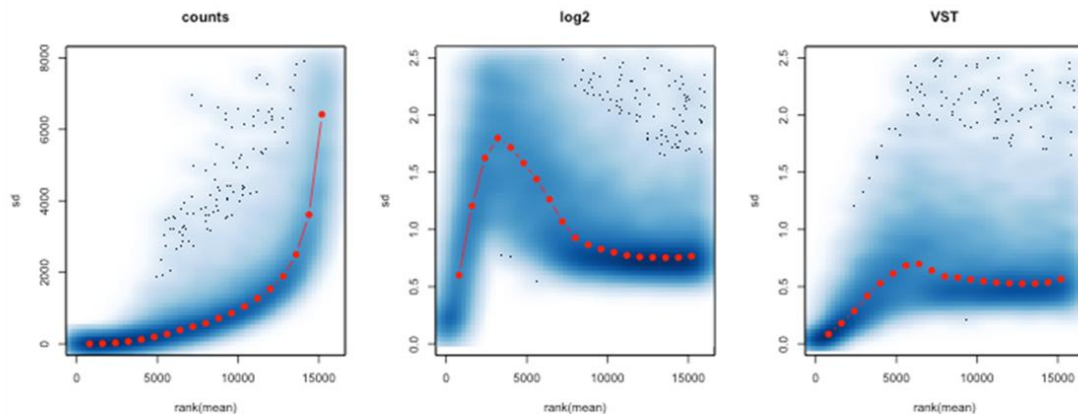
This prevents **high-count variation** from
dominating PCA differences and biological
clustering

Logarithmic Transformation

The simplest and fastest method, intuitive for fold changes. Lowly expressed genes show higher variation.

VST (Variance Stabilizing Transformation)

Results in less variable values across the expression range, preventing low-count variation from affecting downstream analysis



$$y = \log_2(n + n_0)$$

$$\tau(\kappa) = \int^{\kappa} \frac{dq}{\sqrt{w(q)}}.$$

Most network and statistical methods assume roughly symmetric data with stable variance — transformation gives them that.

Gene filtering

Not every measured gene is useful

Remove uninformative genes

- **Low expression**
 - detected in only a handful of samples → mostly noise.
- **Low variance**
 - barely changes across samples → little information about relationships.
- **Typical criteria**
 - keep genes above a minimum expression in a minimum number of samples, or keep the most variable genes.

Why it matters

- **Lowers dimensionality**
 - fewer nodes to connect → eases 'large p, small n'.
- **Cuts spurious edges**
 - noisy, flat genes generate false correlations.
- **Speeds computation**
 - smaller, cleaner matrix to model.

Exploratory analysis



See global structure

Do samples group the way the biology predicts?



Catch technical artifacts

Batch effects hiding as biological signal.



Find outliers

Mislabelled or low-quality samples.



Guide earlier choices

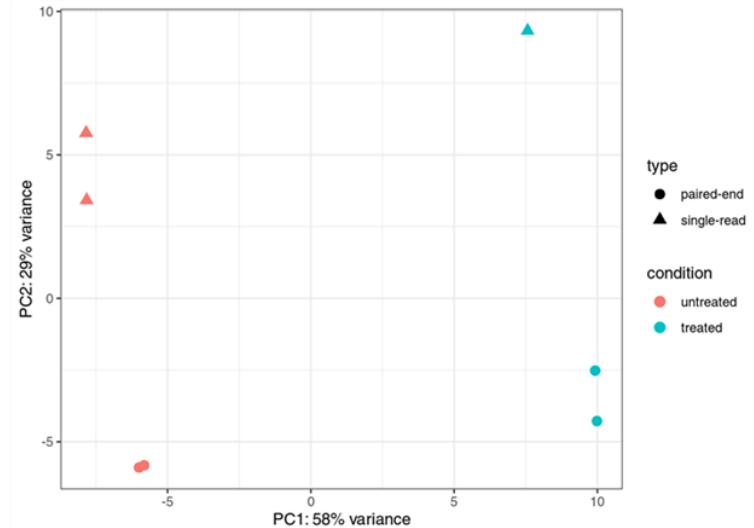
A final check on normalization & filtering.

Exploratory data analysis: PCA

- Are there any **outliers** in my dataset?
- Are samples from same condition/time point **clustered** together?



Principal Component Analysis (**PCA**) reduces data dimensionality by transforming thousands of correlated genes into fewer uncorrelated Principal Components (PCs) that capture **maximum data variance**



Separate cluster = biological differences

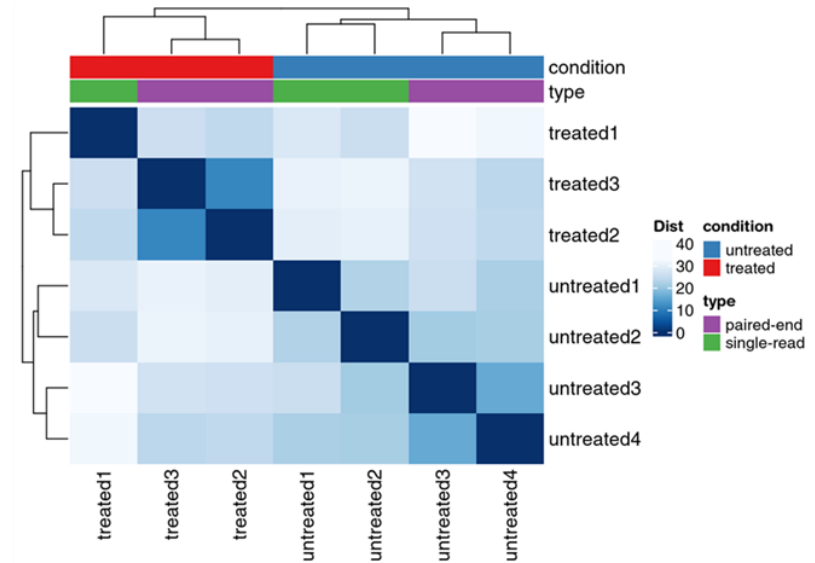
Exploratory data analysis: Clustering

1. Sample Distances

Calculate **euclidean distances** on VST data to quantify **similarity** between samples

1. Heatmap Visualization

Apply **hierarchical clustering** to the distance matrix to identify **outliers** and check replicate **consistency**



How to check and remove confounders factors?

Run PCA



Check for **clustering by metadata** (e.g., Date, Technician, type of sequencing) instead of **biological** condition



Do I see a **known batch effect**?

YES



NO



Run **ComBat-Seq** (**sva** package) to adjust counts for a known **technical** bias

Run **sva** (**sva** package) to identify and remove hidden technical noise or unmeasured artifacts via "surrogate variables"

Key Points

1

Normalization = comparability

Remove depth and composition differences. For RNA-seq: TMM or median-of-ratios.

2

Transformation = stability

Log or VST so variance stops tracking the mean; symmetric data for downstream methods.

3

Filtering = signal

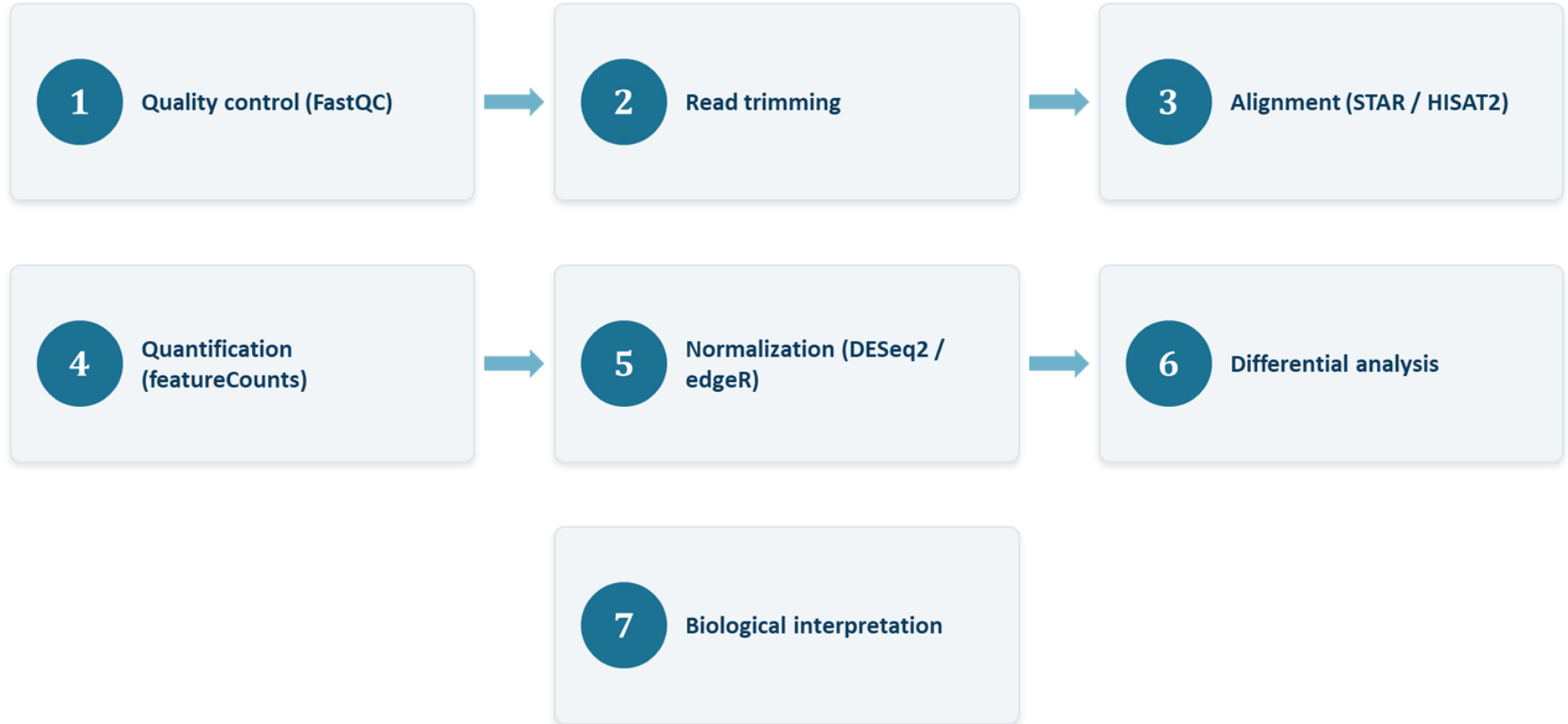
Drop low-expression, low-variance genes: fewer nodes, less noise, fewer false edges.

4

EDA = validation

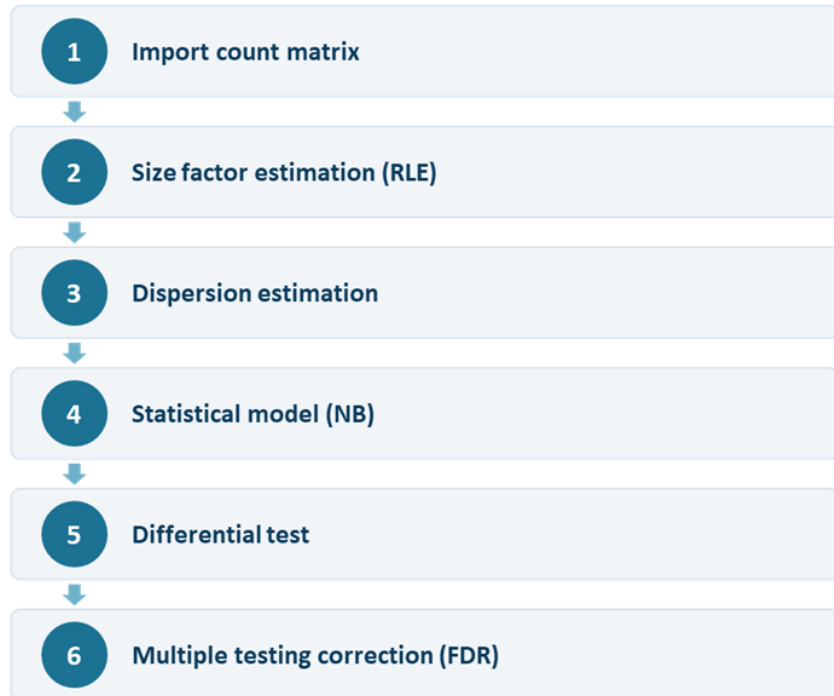
PCA, correlation heatmaps and batch checks confirm structure before modelling.

Bulk RNA-seq workflow

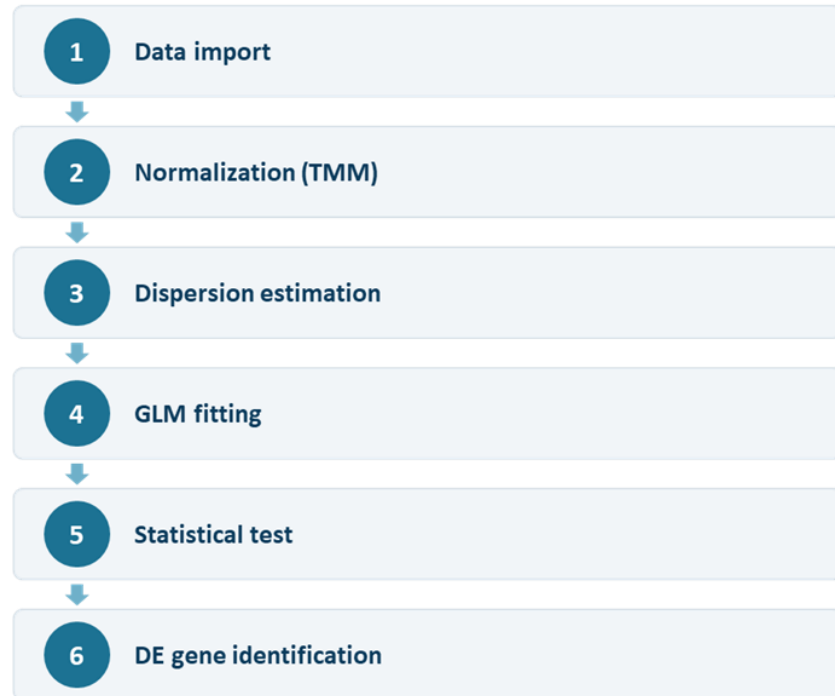


Differential expression workflows: DESeq2 vs edgeR

DESeq2 *RLE normalization*



edgeR *TMM normalization*



A recap on the dataset

We are using the *pasilla* dataset, a Bioconductor package derived from a study on *Drosophila melanogaster*:

- It investigates the effects of **knocking down** the **Pasilla** gene (a splicing factor) at **steady state**
- RNA-seq counts for **7 samples** (4 untreated and 3 treated)
- Metadata information (sequencing type)

